CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 24: Solution of Right Triangles

Topic-wise short notes with solved examples

Finding unknown sides and angles of right-angled triangles using trigonometrical ratios of standard angles, including figures made of two or more right triangles.

1 What "solving" means

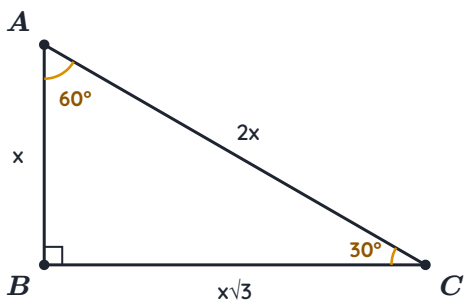
- A triangle has six parts: three sides and three angles. To **solve** a right triangle is to find its unknown parts from the given ones.
- A right triangle can be solved when, besides the right angle, we know **one side and one acute angle**, or **two sides**.
- The two acute angles add up to $90°$.

To find	Known	Use
Perpendicular	Hypotenuse and angle	$\sin\theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$
Base	Hypotenuse and angle	$\cos\theta = \frac{\text{base}}{\text{hypotenuse}}$
Perpendicular	Base and angle	$\tan\theta = \frac{\text{perpendicular}}{\text{base}}$
Angle	Two sides	The ratio of those two sides

- Choose the ratio that connects the **known side** and the **wanted side**.

2 The two special triangles

- **30-60-90 triangle**: sides opposite $30°, 60°, 90°$ are in the ratio $1 : \sqrt{3} : 2$. The side opposite $30°$ is half the hypotenuse.
- **45-45-90 triangle**: sides are in the ratio $1 : 1 : \sqrt{2}$.



A 30-60-90 triangle with sides x, x√3 and 2x

3 One side and one angle given

Example 3.1 In triangle ABC, $\angle B = 90°$, $\angle C = 30°$ and $AC = 20$ cm. Find AB and BC.

SOLUTION

$AB = AC \sin 30° = 20 \times \frac{1}{2}$ and $BC = AC \cos 30° = 20 \times \frac{\sqrt{3}}{2}$

$AB = 10$ cm, $BC = 10\sqrt{3}$ cm

Example 3.2 In triangle PQR, $\angle Q = 90°$, $\angle P = 60°$ and $PQ = 8$ cm. Find QR and PR.

SOLUTION

$\tan 60° = \frac{QR}{8}$ gives $QR = 8\sqrt{3}$. $\cos 60° = \frac{8}{PR}$ gives $PR = 16$.

$QR = 8\sqrt{3}$ cm, $PR = 16$ cm

Example 3.3 The diagonal of a square is $10\sqrt{2}$ cm. Find its side.

SOLUTION

The diagonal makes $45°$ with a side: side $= 10\sqrt{2} \cos 45°$

10 cm

4 Two sides given

Example 4.1 In triangle ABC, $\angle B = 90°$, $AB = 5$ cm and $AC = 10$ cm. Find $\angle A$, $\angle C$ and BC.

SOLUTION

$\cos A = \frac{5}{10} = \frac{1}{2}$, so $\angle A = 60°$ and $\angle C = 30°$.

$BC = 10 \sin 60°$

$\angle A = 60°$, $\angle C = 30°$, $BC = 5\sqrt{3}$ cm

Example 4.2 In triangle ABC, $\angle B = 90°$ and $AB = BC = 7$ cm. Find the angles and AC.

SOLUTION

$\tan A = 1$, so $\angle A = \angle C = 45°$. $AC = 7\sqrt{2}$.

$45°, 45°$; $7\sqrt{2}$ cm

5 Triangles with an altitude

- Any triangle can be split into two right triangles by an altitude. Solve each part and combine.
- The altitude is the common side: find it from one triangle and use it in the other.

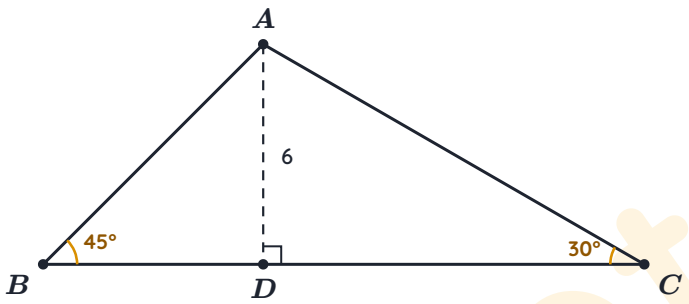
Example 5.1 In triangle ABC, AD is perpendicular to BC, $\angle B = 45^\circ$, $\angle C = 30^\circ$ and $AD = 6$ cm. Find BC.

SOLUTION

In triangle ABD: $\tan 45^\circ = \dfrac{6}{BD}$, so $BD = 6$ cm.

In triangle ADC: $\tan 30^\circ = \dfrac{6}{DC}$, so $DC = 6\sqrt{3}$ cm.

$BC = 6 + 6\sqrt{3} = 6(1 + \sqrt{3})$ **cm, about 16.39 cm**



AD is the common side of the two right triangles

Example 5.2 In triangle ABC, $AB = AC = 10$ cm and $\angle B = 30^\circ$. Find BC and the altitude from A.

SOLUTION

The altitude AD bisects BC. $AD = 10 \sin 30^\circ = 5$ cm and $BD = 10 \cos 30^\circ = 5\sqrt{3}$ cm.

$BC = 10\sqrt{3}$ **cm; altitude 5 cm**

6 Two triangles on a common side

- When two right triangles share a side, set up an equation for the unknown length using the tangent in each.

Example 6.1 From a point C on the ground, the top A of a vertical pole AB is seen at 60°. From D, 20 m farther from the pole on the same line, it is seen at 30°. Find the height of the pole.

SOLUTION

Let $AB = h$ and $BC = x$. Then $h = x\sqrt{3}$ and $h = \dfrac{x + 20}{\sqrt{3}}$.

$3x = x + 20$, so $x = 10$

$h = 10\sqrt{3}$ **m, about 17.32 m**

Example 6.2 In a trapezium ABCD, $AB \parallel DC$, $\angle A = \angle B = 60^\circ$, $DC = 8$ cm and the height is $5\sqrt{3}$ cm. Find AB.

SOLUTION

Drop perpendiculars from D and C. Each end triangle has base $\dfrac{5\sqrt{3}}{\tan 60^\circ} = 5$ cm.

$AB = 8 + 5 + 5 = 18$ **cm**

7 Other figures

Example 7.1 A rhombus has side 12 cm and one angle 60°. Find its diagonals.

SOLUTION

The diagonals bisect the angles and meet at right angles. Half the shorter diagonal $= 12 \sin 30^\circ = 6$ cm; half the longer $= 12 \cos 30^\circ = 6\sqrt{3}$ cm.

12 cm and $12\sqrt{3}$ cm

Example 7.2 A ladder 6 m long leans against a wall and makes 60° with the ground. How high does it reach, and how far is its foot from the wall?

SOLUTION

Height $= 6 \sin 60^\circ$; distance $= 6 \cos 60^\circ$

$3\sqrt{3}$ m (about 5.2 m) and 3 m

Example 7.3 A chord of a circle of radius 10 cm subtends 120° at the centre. Find its length.

SOLUTION

The perpendicular from the centre bisects the chord and the angle: half the chord $= 10 \sin 60^\circ = 5\sqrt{3}$ cm.

$10\sqrt{3}$ cm

8 Common mistakes

- Choosing a ratio that does not include the known side.
- Using $\sin$ where $\cos$ is needed, by mixing up the perpendicular and the base.
- Applying the ratios in a triangle that has no right angle. Draw an altitude first.
- Writing $\tan 30^\circ = \sqrt{3}$.
- Forgetting to add (or subtract) the two parts of the base at the end.
- Rounding too early: keep surds until the last step. Use $\sqrt{3} \approx 1.732$ and $\sqrt{2} \approx 1.414$.